

Lecture 08: Labor Supply and Taxable Income – Live In-Class Empirical Exercise

Stata Code (Uncommented Section Only)

```
*****
* Figure 4 replication (Kleven-Landaïs-Saez AER)
* Denmark tax reform + Bosman ruling
*****
*
* Why split into "sports/entertainment" vs "all other industries"?
* 1) Mobility differs by sector: sports workers (incl. football) are highly
   mobile.
* 2) 1995 Bosman ruling is a sector-specific shock for football labor markets.
* 3) The non-sports panel is a comparison panel to assess whether patterns are
   specific to sports or visible more broadly around the tax reform.
*
* Main confounds to keep in mind:
* - Bosman ruling in 1995 (major confound for sports panel after 1995).
* - Differential sectoral demand shocks over time (sports vs non-sports trends
  ).
* - Cross-country macro shocks and league-specific institutional changes.
* - Composition changes in who is above/below the eligibility threshold.
* - Normalization to 1990 helps comparability but does not remove all
  confounds.
*
version 13.1
clear all
cap log close
set more off

* Paths
if "`c(username)'" == "muhammadbashir" {
    global INCLASS_DIR "/Users/muhammadbashir/Library/CloudStorage/Dropbox/
    /ugradteaching/econ131spring26/LectureEmpiricalExercises/
    Lecture08_laborsupplyandtaxableincome/inclass"
    global SOURCE_DIR "/Users/muhammadbashir/Library/CloudStorage/Dropbox/
    /ugradteaching/econ131spring26/LectureEmpiricalExercises/
    Lecture08_laborsupplyandtaxableincome/kleven-landais-
    saezAER13football/Data/_Prog_AER"
}
else if "`c(username)'" == "yagan" {
    global INCLASS_DIR "C:/Users/yagan/Dropbox/ugradteaching/
    econ131spring26/LectureEmpiricalExercises/
    Lecture08_laborsupplyandtaxableincome/inclass"
```

```

        global SOURCE_DIR "C:/Users/yagan/Dropbox/ugradteaching/
        econ131spring26/LectureEmpiricalExercises/
        Lecture08_laborsupplyandtaxableincome/kleven-landais-
        saezAER13football/Data_&_Prog_AER"
    }
    global DATA_DIR "$SOURCE_DIR/data/data_graph"
    global OUT_DIR "$INCLASS_DIR"

    cd "$INCLASS_DIR"
    log using "$OUT_DIR/replicate_figure4_denmark_taxscheme_output.log", text
    replace

    * -----
    * Panel A: Sports + entertainment (culture1)
    * -----
    *(1) SET UP THE DATA
    insheet using "$DATA_DIR/aer_culture1.csv", clear
    * In released replication data, tr is already coded:
    * tr=1 treatment (above eligibility threshold), tr=0 control (80%-99.5% of
      threshold).
    keep if inlist(tr,0,1) & year<=2005
    * Rebuild plotted series from counts:
    * count_workers = raw number of foreign workers in each group-year
    * count_norm = normalized to 1 in 1990 within tr group
    gen count_workers = foreigner2
    gen base1990 = count_workers if year==1990
    bys tr: egen base1990_tmp = max(base1990)
    drop base1990
    rename base1990_tmp base1990
    gen count_norm = count_workers/base1990
    gen post_reform = year>=1991
    sort tr year

    *(2) PLOT THE RAW DATA BY TREATMENT AND CONTROL YEAR-BY-YEAR
    twoway ///
        (connected count_workers year if tr==0, ///
            msymbol(T) mcolor(cranberry) lcolor(cranberry) lpattern(dash)) ///
        (connected count_workers year if tr==1, ///
            msymbol(O) mcolor(navy) lcolor(navy) lpattern(solid)), ///
    graphregion(fcolor(white)) ///
    xlabel(1980(5)2005, grid glc(gs14)) ///
    xline(1991, lcolor(red) lpattern(solid)) ///
    xline(1995, lcolor(red) lpattern(dash)) ///
    legend(order(1 "Control" 2 "Treatment") col(2) ring(0) pos(11)) ///
    xtitle("Year") ///
    ytitle("Raw count of foreign workers") ///
    ttext(122 1984.7 "Before the reform", just(left) size(small)) ///
    ttext(122 1992.7 "After reform" "before Bosman ruling", just(left) size(
        small)) ///
    ttext(122 1997.6 "After reform" "after Bosman ruling", just(left) size(
        small)) ///
    title("Panel A. Sports and entertainment (raw counts)", size(medsmall))
    ///

```

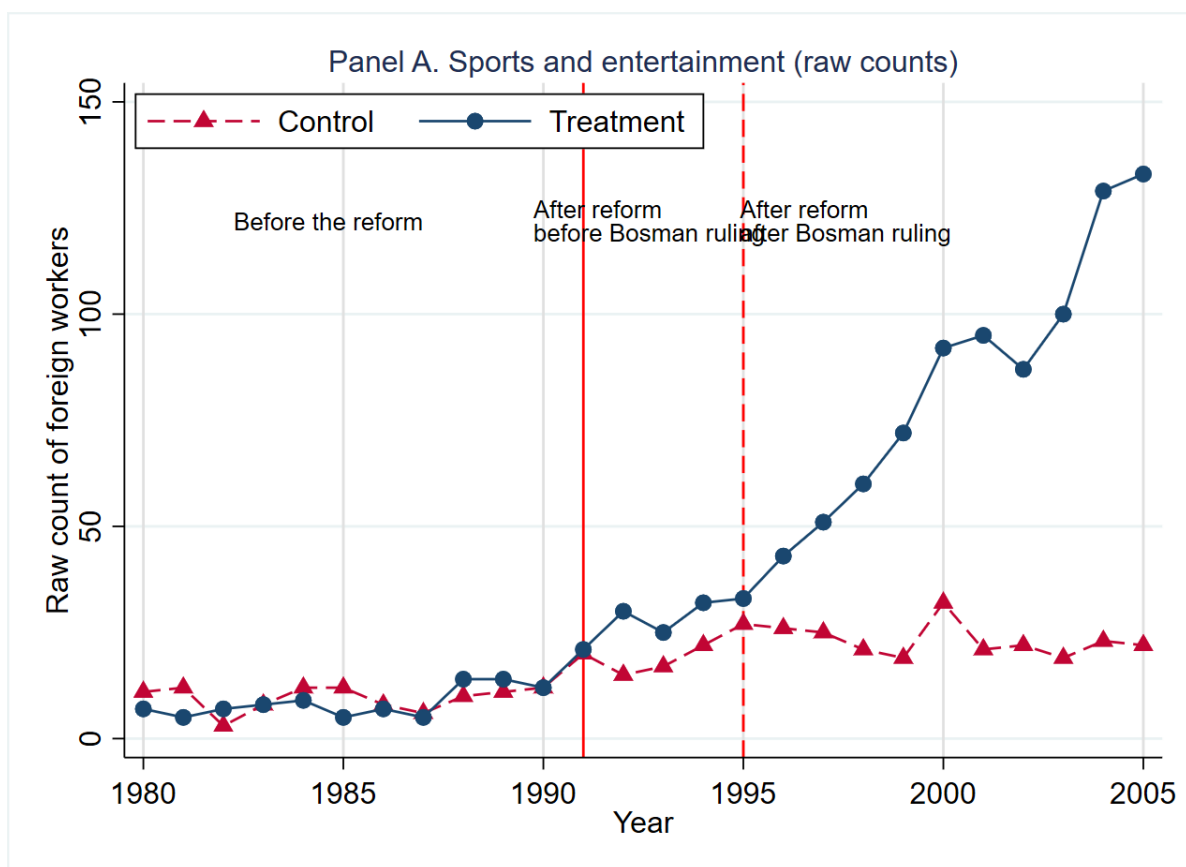
```

name(fig4_panelA_raw, replace)

*(3) CONSTRUCT THE DD TABLE
* DiD estimates for Panel A (raw counts only):
su count_workers if tr==1 & post_reform==1
scalar mt_post = r(mean)
su count_workers if tr==1 & post_reform==0
scalar mt_pre = r(mean)
su count_workers if tr==0 & post_reform==1
scalar mc_post = r(mean)
su count_workers if tr==0 & post_reform==0
scalar mc_pre = r(mean)
scalar diff_treat = mt_post-mt_pre
scalar diff_control = mc_post-mc_pre
scalar did_mean = diff_treat - diff_control
di      "DiD estimate (mean comparison) for Panel A: " %9.8g did_mean

```

Figure 4, Panel A: Sports and Entertainment (Raw Counts)



Stata Output (di + Regression)

```
-----
      name:  <unnamed>
      log:   /Users/muhammadbashir/Library/CloudStorage/Dropbox/ugradteaching/e
> con131spring26/LectureEmpiricalExercises/Lecture08_laborsupplyandtaxableincom
> e/inclass/replicate_figure4_denmark_taxescheme_output.log
      log type:  text
      opened on:  11 Mar 2026, 10:25:10

.
. * -----
. * Panel A: Sports + entertainment (culture1)
. * -----
. *(1) SET UP THE DATA
. insheet using "$DATA_DIR/aer_culture1.csv", clear
(17 vars, 84 obs)

. * In released replication data, tr is already coded:
. * tr=1 treatment (above eligibility threshold), tr=0 control (80%-99.5% of th
> reshold).
. keep if inlist(tr,0,1) & year<=2005
(32 observations deleted)

. * Rebuild plotted series from counts:
. * count_workers = raw number of foreign workers in each group-year
. * count_norm     = normalized to 1 in 1990 within tr group
. gen count_workers = foreigner2

. gen base1990 = count_workers if year==1990
(50 missing values generated)

. bys tr: egen base1990_tmp = max(base1990)

. drop base1990

. rename base1990_tmp base1990

. gen count_norm = count_workers/base1990

. gen post_reform = year>=1991

. sort tr year

.
.
. *(2) PLOT THE RAW DATA BY TREATMENT AND CONTROL YEAR-BY-YEAR
. twoway ///
>   (connected count_workers year if tr==0, ///
>     msymbol(T) mcolor(cranberry) lcolor(cranberry) lpattern(dash)) ///
>   (connected count_workers year if tr==1, ///
>     msymbol(O) mcolor(navy) lcolor(navy) lpattern(solid)), ///
>   graphregion(fcolor(white)) ///
>   xlabel(1980(5)2005, grid glc(gs14)) ///
>   xline(1991, lcolor(red) lpattern(solid)) ///
>   xline(1995, lcolor(red) lpattern(dash)) ///
>   legend(order(1 "Control" 2 "Treatment") col(2) ring(0) pos(11)) ///
>   xtitle("Year") ///
>   ytitle("Raw count of foreign workers") ///
>   ttext(122 1984.7 "Before the reform", just(left) size(small)) ///
>   ttext(122 1992.7 "After reform" "before Bosman ruling", just(left) size(s
> mall)) ///
>   ttext(122 1997.6 "After reform" "after Bosman ruling", just(left) size(sm
> all)) ///
>   title("Panel A. Sports and entertainment (raw counts)", size(medsmall)) /
> //
>   name(fig4_panelA_raw, replace)
```

```

.
.
. *(3) CONSTRUCT THE DD TABLE
. * DiD estimates for Panel A (raw counts only):
. su count_workers if tr==1 & post_reform==1

      Variable |           Obs       Mean   Std. dev.       Min       Max
-----+-----
count_work~s |           15   66.86667   37.44876         21       133

. scalar mt_post = r(mean)

. su count_workers if tr==1 & post_reform==0

      Variable |           Obs       Mean   Std. dev.       Min       Max
-----+-----
count_work~s |           11    8.454545   3.416537         5        14

. scalar mt_pre = r(mean)

. su count_workers if tr==0 & post_reform==1

      Variable |           Obs       Mean   Std. dev.       Min       Max
-----+-----
count_work~s |           15   22.06667   4.216747        15        32

. scalar mc_post = r(mean)

. su count_workers if tr==0 & post_reform==0

      Variable |           Obs       Mean   Std. dev.       Min       Max
-----+-----
count_work~s |           11    9.545455   2.978712         3        12

. scalar mc_pre = r(mean)

. scalar diff_treat = mt_post-mt_pre

. scalar diff_control = mc_post-mc_pre

. scalar did_mean = diff_treat - diff_control

. di      "DiD estimate (mean comparison) for Panel A: " %9.8g did_mean
DiD estimate (mean comparison) for Panel A:  45.89091

.
.
. *REPLICATE USING REGRESSION
. * Manual DiD dummy construction
. gen D_treat = (tr==1)

. gen D_post = (post_reform==1)

. gen D_treat_post = D_treat*D_post

. reg count_workers D_treat D_post D_treat_post, vce(robust)

Linear regression              Number of obs   =          52
                               F(3, 48)       =         44.66
                               Prob > F        =         0.0000
                               R-squared        =         0.6070
                               Root MSE     =         20.457

-----+-----
count_work~s |           Robust
              | Coefficient std. err.      t    P>|t|    [95% conf. interval]
-----+-----

```

D_treat		-1.090909	1.35627	-0.80	0.425	-3.817873	1.636055
D_post		12.52121	1.411722	8.87	0.000	9.682755	15.35967
D_treat_post		45.89091	9.877795	4.65	0.000	26.03027	65.75155
_cons		9.545455	.8912857	10.71	0.000	7.753405	11.3375

```

. log close
  name: <unnamed>
  log: /Users/muhammadbashir/Library/CloudStorage/Dropbox/ugradteaching/e
> con131spring26/LectureEmpiricalExercises/Lecture08_laborsupplyandtaxableincom
> e/inclass/replicate_figure4_denmark_taxscheme_output.log
  log type: text
  closed on: 11 Mar 2026, 10:25:10

```
